

Salivary Secretory Immunoglobulin a secretion increases after 4-weeks ingestion of chlorella-derived multicomponent supplement in humans: a randomized cross over study

Background Chlorella, a unicellular green alga that grows in fresh water, contains high levels of proteins, vitamins, minerals, and dietary fibers. Some studies have reported favorable immune function-related effects on biological secretions such as blood and breast milk in humans who have ingested a chlorella-derived multicomponent supplement. However, the effects of chlorella-derived supplement on mucosal immune functions remain unclear. The purpose of this study was to investigate whether chlorella ingestion increases the salivary secretory immunoglobulin A (SIgA) secretion in humans using a blind, randomized, crossover study design. **Methods** Fifteen men took 30 placebo and 30 chlorella tablets per day for 4 weeks separated by a 12-week washout period. Before and after each trial, saliva samples were collected from a sterile cotton ball that was chewed after overnight fasting. Salivary SIgA concentrations were measured using ELISA. **Results** Compliance rates for placebo and chlorella ingestions were $97.0 \pm 1.0\%$ and $95.3 \pm 1.6\%$, respectively. No difference was observed in salivary SIgA concentrations before and after placebo ingestion ($P = 0.38$). However, salivary SIgA concentrations were significantly elevated after chlorella ingestion compared to baseline ($P < 0.01$). No trial $\times$ period interaction was identified for the saliva flow rates. Although the SIgA secretion rate was not affected by placebo ingestion ($P = 0.36$), it significantly increased after 4-week chlorella ingestion than before intake ($P < 0.01$). **Conclusions** These results suggest 4-week ingestion of a chlorella-derived multicomponent supplement increases salivary SIgA secretion and possibly improves mucosal immune function in humans.